

## TECAFORM AD black - Stock Shapes (rods, plates, tubes)

### Chemical Designation

POM-H (Polyacetal (Homopolymer))

### Colour

black opaque

### Density

1.43 g/cm<sup>3</sup>

### Main features

- high strength
- difficult to bond
- good slide and wear properties
- good machinability
- not hot water resistant over 60°C
- good chemical resistance
- easy to polish

### Target Industries

- mechanical engineering
- aircraft and aerospace technology
- electronics
- oil and gas industry
- automotive industry

| Mechanical properties                 | parameter                                   | value            | unit                             | norm                 | comment                                                                                                                                                                                         |
|---------------------------------------|---------------------------------------------|------------------|----------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tensile strength                      | 50mm/min                                    | 80               | MPa                              | DIN EN ISO 527-2     | (1) For tensile test: specimen type 1b                                                                                                                                                          |
| Modulus of elasticity (tensile test)  | 1mm/min                                     | 3600             | MPa                              | DIN EN ISO 527-2     | (2) For flexural test: support span 64mm, norm specimen.                                                                                                                                        |
| Tensile strength at yield             | 50mm/min                                    | 80               | MPa                              | DIN EN ISO 527-2     | (3) Specimen 10x10x10mm                                                                                                                                                                         |
| Elongation at yield (tensile test)    | 50mm/min                                    | 32               | %                                | DIN EN ISO 527-2     | (4) Specimen 10x10x50mm, modulus range between 0.5 and 1% compression.                                                                                                                          |
| Elongation at break (tensile test)    | 50mm/min                                    | 43               | %                                | DIN EN ISO 527-2     | (5) For Charpy test: support span 64mm, norm specimen.                                                                                                                                          |
| Flexural strength                     | 2mm/min, 10 N                               | 106              | MPa                              | DIN EN ISO 178       | n.b. = not broken                                                                                                                                                                               |
| Modulus of elasticity (flexural test) | 2mm/min, 10 N                               | 3600             | MPa                              | DIN EN ISO 178       | (6) Specimen in 4mm thickness                                                                                                                                                                   |
| Compression strength                  | 1% / 2% / 5%<br>5mm/min, 10 N               | 22/38/72         | MPa                              | EN ISO 604           | (3)                                                                                                                                                                                             |
| Compression modulus                   | 5mm/min, 10 N                               | 2800             | MPa                              | EN ISO 604           | (4)                                                                                                                                                                                             |
| Impact strength (Charpy)              | max. 7,5J                                   | n.b.             | kJ/m <sup>2</sup>                | DIN EN ISO 179-1eU   | (5)                                                                                                                                                                                             |
| Notched impact strength (Charpy)      | max. 7,5J                                   | 14               | kJ/m <sup>2</sup>                | DIN EN ISO 179-1eA   |                                                                                                                                                                                                 |
| Ball indentation hardness             |                                             | 185              | MPa                              | ISO 2039-1           | (6)                                                                                                                                                                                             |
| Thermal properties                    | parameter                                   | value            | unit                             | norm                 | comment                                                                                                                                                                                         |
| Glass transition temperature          |                                             | -60              | °C                               | DIN EN ISO 11357     | (1) Found in public sources.                                                                                                                                                                    |
| Melting temperature                   |                                             | 182              | °C                               | DIN EN ISO 11357     | (2) Found in public sources.                                                                                                                                                                    |
| Service temperature                   | short term                                  | 150              | °C                               |                      | Individual testing regarding application conditions is mandatory.                                                                                                                               |
| Service temperature                   | long term                                   | 110              | °C                               |                      |                                                                                                                                                                                                 |
| Thermal expansion (CLTE)              | 23-60°C, long.                              | 11               | 10 <sup>-5</sup> K <sup>-1</sup> | DIN EN ISO 11359-1;2 |                                                                                                                                                                                                 |
| Thermal expansion (CLTE)              | 23-100°C, long.                             | 11               | 10 <sup>-5</sup> K <sup>-1</sup> | DIN EN ISO 11359-1;2 |                                                                                                                                                                                                 |
| Specific heat                         |                                             | 1.3              | J/(g*K)                          | ISO 22007-4:2008     |                                                                                                                                                                                                 |
| Thermal conductivity                  |                                             | 0.43             | W/(K*m)                          | ISO 22007-4:2008     |                                                                                                                                                                                                 |
| Electrical properties                 | parameter                                   | value            | unit                             | norm                 | comment                                                                                                                                                                                         |
| surface resistivity                   | Silver electrode, 23°C, 12% r.h.            | 10 <sup>14</sup> | Ω                                | DIN IEC 60093        | (1) Specimen in 20mm thickness                                                                                                                                                                  |
| volume resistivity                    | Silver electrode, 23°C, 12% r.h.            | 10 <sup>14</sup> | Ω*cm                             | DIN IEC 60093        | (2) Due to the black colourant and moisture uptake of the material the electrical insulation properties cannot be 100% guaranteed, despite single measurements suggesting otherwise.            |
| Dielectric strength                   | 23°C, 50% r.h.                              | 38               | kV/mm                            | ISO 60243-1          | (3) Specimen in 1mm thickness                                                                                                                                                                   |
| Resistance to tracking (CTI)          | Platin electrode, 23°C, 50% r.h., solvent A | 600              | V                                | DIN EN 60112         |                                                                                                                                                                                                 |
| Other properties                      | parameter                                   | value            | unit                             | norm                 | comment                                                                                                                                                                                         |
| Water absorption                      | 24h / 96h (23°C)                            | 0.05 / 0.1       | %                                | DIN EN ISO 62        | (1) Ø ca. 50mm, h=13mm                                                                                                                                                                          |
| Resistance to hot water/ bases        |                                             | -                |                                  | -                    | (2) - poor resistance                                                                                                                                                                           |
| Resistance to weathering              |                                             | -                |                                  | -                    | (3) Corresponding means no listing at UL (yellow card). The information might be taken from resin, stock shape or estimation. Individual testing regarding application conditions is mandatory. |
| Flammability (UL94)                   | corresponding to                            | HB               |                                  | DIN IEC 60695-11-10; | (3)                                                                                                                                                                                             |

Our information and statements reflect the current state of our knowledge and shall inform about our products and their applications. They do not assure or guarantee chemical resistance, quality of products and their merchantability in a legally binding way. Our products are not defined for use in medical or dental implants. Existing commercial patents have to be observed. The corresponding values and information are no minimum or maximum values, but guideline values that can be used primarily for comparison purposes for material selection. These values are within the normal tolerance range of product properties and do not represent guaranteed property values. Therefore they shall not be used for specification purposes. Unless otherwise noted, these values were determined by tests at reference dimensions (typically rods with diameter 40-60 mm according to DIN EN 15860) on extruded and machined specimen. As the properties depend on the dimensions of the semi-finished products and the orientation in the component (esp. in reinforced grades), the material may not be used without a separate testing under individual circumstances. The customer is solely responsible for the quality and suitability of products for the application and has to test usage and processing prior to use.



TECAFORM AD natural - Stock Shapes (rods, plates, tubes)

Chemical Designation

POM-H (Polyacetal (Homopolymer))

Colour

white opaque

Density

1.43 g/cm<sup>3</sup>

Main features

- good slide and wear properties
- high strength
- electrically insulating
- good chemical resistance
- difficult to bond
- good machinability
- easy to polish
- not hot water resistant over 60°C

Target Industries

- mechanical engineering
- aircraft and aerospace technology
- electronics
- food technology
- automotive industry

| Mechanical properties                 | parameter                     | value            | unit                             | norm                 | comment                                                                                                                                                                                                                                                                                                              |
|---------------------------------------|-------------------------------|------------------|----------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tensile strength                      | 50mm/min                      | 79               | MPa                              | DIN EN ISO 527-2     | (1) For tensile test: specimen type 1b<br>(2) For flexural test: support span 64mm, norm specimen.<br>(3) Specimen 10x10x10mm<br>(4) Specimen 10x10x50mm, modulus range between 0.5 and 1% compression.<br>(5) For Charpy test: support span 64mm, norm specimen. n.b. = not broken<br>(6) Specimen in 4mm thickness |
| Modulus of elasticity (tensile test)  | 1mm/min                       | 3400             | MPa                              | DIN EN ISO 527-2     |                                                                                                                                                                                                                                                                                                                      |
| Tensile strength at yield             | 50mm/min                      | 79               | MPa                              | DIN EN ISO 527-2     |                                                                                                                                                                                                                                                                                                                      |
| Elongation at yield (tensile test)    | 50mm/min                      | 37               | %                                | DIN EN ISO 527-2     |                                                                                                                                                                                                                                                                                                                      |
| Elongation at break (tensile test)    | 50mm/min                      | 45               | %                                | DIN EN ISO 527-2     |                                                                                                                                                                                                                                                                                                                      |
| Flexural strength                     | 2mm/min, 10 N                 | 106              | MPa                              | DIN EN ISO 178       |                                                                                                                                                                                                                                                                                                                      |
| Modulus of elasticity (flexural test) | 2mm/min, 10 N                 | 3600             | MPa                              | DIN EN ISO 178       |                                                                                                                                                                                                                                                                                                                      |
| Compression strength                  | 1% / 2% / 5%<br>5mm/min, 10 N | 19/33/69         | MPa                              | EN ISO 604           |                                                                                                                                                                                                                                                                                                                      |
| Compression modulus                   | 5mm/min, 10 N                 | 2700             | MPa                              | EN ISO 604           |                                                                                                                                                                                                                                                                                                                      |
| Impact strength (Charpy)              | max. 7,5J                     | n.b.             | kJ/m <sup>2</sup>                | DIN EN ISO 179-1eU   |                                                                                                                                                                                                                                                                                                                      |
| Notched impact strength (Charpy)      | max. 7,5J                     | 15               | kJ/m <sup>2</sup>                | DIN EN ISO 179-1eA   |                                                                                                                                                                                                                                                                                                                      |
| Ball indentation hardness             |                               | 185              | MPa                              | ISO 2039-1           | (6)                                                                                                                                                                                                                                                                                                                  |
| Thermal properties                    | parameter                     | value            | unit                             | norm                 | comment                                                                                                                                                                                                                                                                                                              |
| Glass transition temperature          |                               | -60              | °C                               | DIN EN ISO 11357     | (1) Found in public sources.                                                                                                                                                                                                                                                                                         |
| Melting temperature                   |                               | 182              | °C                               | DIN EN ISO 11357     | (2) Found in public sources.                                                                                                                                                                                                                                                                                         |
| Service temperature                   | short term                    | 150              | °C                               |                      | Individual testing regarding application conditions is mandatory.                                                                                                                                                                                                                                                    |
| Service temperature                   | long term                     | 110              | °C                               |                      |                                                                                                                                                                                                                                                                                                                      |
| Thermal expansion (CLTE)              | 23-60°C, long.                | 12               | 10 <sup>-5</sup> K <sup>-1</sup> | DIN EN ISO 11359-1;2 |                                                                                                                                                                                                                                                                                                                      |
| Thermal expansion (CLTE)              | 23-100°C, long.               | 13               | 10 <sup>-5</sup> K <sup>-1</sup> | DIN EN ISO 11359-1;2 |                                                                                                                                                                                                                                                                                                                      |
| Specific heat                         |                               | 1.3              | J/(g*K)                          | ISO 22007-4:2008     |                                                                                                                                                                                                                                                                                                                      |
| Thermal conductivity                  |                               | 0.43             | W/(K*m)                          | ISO 22007-4:2008     |                                                                                                                                                                                                                                                                                                                      |
| Electrical properties                 | parameter                     | value            | unit                             | norm                 | comment                                                                                                                                                                                                                                                                                                              |
| surface resistivity                   |                               | 10 <sup>14</sup> | Ω                                | DIN IEC 60093        |                                                                                                                                                                                                                                                                                                                      |
| Other properties                      | parameter                     | value            | unit                             | norm                 | comment                                                                                                                                                                                                                                                                                                              |
| Water absorption                      | 24h / 96h (23°C)              | 0.05 / 0.1       | %                                | DIN EN ISO 62        | (1) Ø ca. 50mm, h=13mm                                                                                                                                                                                                                                                                                               |
| Resistance to hot water/ bases        |                               | -                |                                  | -                    | (2) - poor resistance                                                                                                                                                                                                                                                                                                |
| Resistance to weathering              |                               | -                |                                  | -                    | (3) Corresponding means no listing at UL (yellow card). The information might be taken from resin, stock shape or estimation. Individual testing regarding application conditions is mandatory.                                                                                                                      |
| Flammability (UL94)                   | corresponding to              | HB               |                                  | DIN IEC 60695-11-10; | (3)                                                                                                                                                                                                                                                                                                                  |

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